

Struktur Project

```
svm-fastapi-streamlit/  
|  
├─ train_model.ipynb      # Google Colab  
├─ model.pkl  
├─ scaler.pkl  
├─ api.py                 # FastAPI  
├─ app.py                 # Streamlit  
├─ requirements.txt  
├─ Procfile  
├─ runtime.txt  
├─ .gitignore  
└─ README.md
```

1. Google Colab (train_model.ipynb)

Dataset menggunakan Breast Cancer dari sklearn.

</> Python

```
# =====  
# IMPORT LIBRARY  
# =====  
  
import pandas as pd  
import numpy as np  
  
from sklearn.datasets import load_breast_cancer  
from sklearn.model_selection import train_test_split  
from sklearn.preprocessing import StandardScaler  
from sklearn.svm import SVC  
from sklearn.metrics import accuracy_score  
import pickle
```

2. Download Pickle

</> Python

```
from google.colab import files  
  
files.download("model.pkl")  
files.download("scaler.pkl")
```

3. FastAPI (api.py)

Python

```
from fastapi import FastAPI
from pydantic import BaseModel

import pickle
import numpy as np

app = FastAPI()

model = pickle.load(open("model.pkl", "rb"))
scaler = pickle.load(open("scaler.pkl", "rb"))

class InputData(BaseModel):

    features: list

@app.get("/")
def home():

    return {"status": "SVM API Running"}
```

Menjalankan FastAPI

Bash

```
uvicorn api:app --reload
```

API akan berjalan di

```
http://127.0.0.1:8000
```

Swagger

```
http://127.0.0.1:8000/docs
```

4. Streamlit (app.py)

Python

```
import streamlit as st
import requests

st.set_page_config(
    page_title="SVM Prediction",
    layout="wide"
)

st.title("Breast Cancer Prediction")

st.write("Input 30 Feature")

inputs=[]

for i in range(30):

    value=st.number_input(f"Feature {i+1}")

    inputs.append(value)

if st.button("Predict"):

    url="http://127.0.0.1:8000/predict"
```

Jika FastAPI sudah di Render misalnya

Python

```
url="https://xxxx.onrender.com/predict"
```

cukup ganti URL tersebut.

5. requirements.txt

```
fastapi
uvicorn
streamlit
scikit-learn
numpy
pandas
requests
pydantic
joblib
```

6. Procfile

```
web: uvicorn api:app --host=0.0.0.0 --port=$PORT
```

7. runtime.txt

```
python-3.11.9
```

8. .gitignore

```
__pycache__/  
*.pyc  
.ipynb_checkpoints/  
  
.env  
  
venv/  
env/  
  
.DS_Store
```

9. Menjalankan Lokal

Instal dependensi:

❏ Bash

```
pip install -r requirements.txt
```

Jalankan FastAPI:

❏ Bash

```
uvicorn api:app --reload
```

Buka terminal baru dan jalankan Streamlit:

❏ Bash

```
streamlit run app.py
```

10. Deployment

Repository GitHub

```
svm-fastapi-streamlit/  
  api.py  
  app.py  
  model.pkl  
  scaler.pkl  
  requirements.txt  
  Procfile  
  runtime.txt
```

Deploy FastAPI

- Push ke GitHub.
- Deploy ke layanan seperti Render atau Railway.
- Salin URL API yang dihasilkan.

Deploy Streamlit

- Login ke Streamlit Community Cloud.
- Hubungkan ke repository GitHub.
- Pilih `app.py` sebagai entry point.
- Ganti URL FastAPI pada `app.py` menjadi URL hasil deployment FastAPI.
- Deploy.

Arsitektur akhirnya:

